

Publication Batch — ARK-493 → ARK-498 Series

Prepared for: Remnant Fieldworks Inc. — ExecutionProof program

Source of truth: this series' run artifacts (`results/` , `proofrecords/` , `ledgers/`) + reconciled corpus totals in `reconciliation_report.md`

Framing rule: every public number below is the **reconciled post-series total**; negative results (FAIL, GATE-STOP) are reported, not suppressed.

Repository count (corrected): this series was published as the `ark-493-498/` subdirectory of the existing `executionproof-testbeds` repository — it creates **no new repository**, so the repository total stays **9** (not 10).

Reconciled public headline numbers (use everywhere):

72 preregistered experiments · 232 case records · 229 PASS · 2 FAIL · 1 GATE-STOP · 9 repositories.

This series (ARK-493-498) contributed 6 experiments and 161 scored cases, all PASS, with 0 new FAIL and 0 GATE-STOP.

1. GitHub README text (drop-in section)

ExecutionProof – Enterprise Agent Boundary Testbed (ARK-493 → ARK-498)

ExecutionProof enforces ****proof before power****: an autonomous agent's tool call executes only after an independent gate has verified actor identity, live authority, evidence, policy version, system state, and exact-action integrity – and every decision is written to a signed, hash-chained, independently reconstructable ProofRecord. Fail-closed is the default; failures are preserved, never retried until they pass.

This series at a glance (frozen preregistration v1.1)

Experiment	Focus	Result
ARK-493	Enforcement boundary under adversarial load (6 paths × 15 cells)	PASS – 90/90
ARK-494	Semantic boundary: deep argument mutation	PASS – 13/13
ARK-495	Temporal boundary: authority change mid-flight	PASS – 11/11
ARK-496	Multi-agent delegation & self-approval defense	PASS – 8/8
ARK-497	Independently reconstructable ProofRecord (+ tamper detection)	PASS – 30/30
ARK-498	Networked production-like performance	PASS – 6/6 hard criteria

- ****161 scored cases this series, 161 PASS, 0 FAIL, 0 enforcement leaks.****
- ****Dual-guard verification:**** an in-process Guard-A and an **isolated-subprocess** Guard-B independently verified all 161 scored records and agreed on every one.
- ****ARK-497:**** an isolated verifier (permitted imports only, statically proven to import nothing from the application) reconstructed all 9 decision elements for 20 legitimate records and detected ****10/10**** single-field tamper attempts with ****0**** false positives.
- ****ARK-498:**** behind a real HTTP/loopback-TCP boundary (~1,810 requests), the gate held fail-closed under policy/authority/store failure (****leak count 0****), executed ****zero**** duplicate side effects, produced ****100%**** complete and signature-verifiable ProofRecords, and recovered cleanly with ****no**** automatic re-execution of denied requests.

Standing corpus (reconciled, all series to date)

****72** preregistered experiments · 232 case records · 229 PASS · 2 FAIL · 1 GATE-STOP · 9 repositories.****** The 2 FAIL and 1 GATE-STOP records are retained unchanged – preserving negative results is part of the methodology.

Reproduce

```
```bash
pip install -r requirements.txt
python3 run_all.py # verifies the frozen preregistration hash, then runs 493→498
```

Preregistration: `preregistration/ARK-493-498-PREREGISTRATION-v1.1.md`

SHA-256 `464b9fb8be9d6cca052f236dc9deec9f8e89b781cafc58701e79b2d05d52952a` .

...

**Performance note for the README/badge:** ARK-498 latency/throughput figures are labeled PRODUCTION-LIKE OVERHEAD CHARACTERIZATION · NOT A BENCHMARK CERTIFICATION · NOT A PRODUCTION SLA. Do not publish them as an SLA.

## 2. Zenodo deposition draft

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**Title:** ExecutionProof Enterprise Agent Boundary Testbed — ARK-493 through ARK-498 (Frozen Preregistration v1.1)

**Authors:** Remnant Fieldworks Inc., ExecutionProof Program

**Version:** 1.0 (series ARK-493–498)

**Resource type:** Software / Dataset (preregistered experiment suite + evidence artifacts)

**Description:**

This deposition contains the frozen preregistration (v1.1), the complete testbed source, and the full evidence artifacts for six preregistered experiments (ARK-493 through ARK-498) characterizing the enforcement boundary of the ExecutionProof agent-authorization gate. The gate verifies six dimensions (actor identity, live authority, evidence, policy version, system state, exact-action integrity) and emits ALLOW / DENY / HOLD, recording every decision in a signed, hash-chained, independently reconstructable ProofRecord.

Results (single reproducible run, preregistration hash verified at execution): 161 scored cases, 161 PASS, 0 FAIL, 0 enforcement leaks; dual-guard (in-process + isolated-subprocess) agreement on all 161 records. ARK-497 demonstrates independent reconstruction of all nine decision elements and 10/10 single-field tamper detection with zero false positives by a verifier statically proven to import nothing from the application. ARK-498 characterizes production-like overhead behind a real HTTP/loopback-TCP boundary (~1,810 requests) and meets all six frozen hard criteria: fail-closed leak count 0, zero duplicate executions, 100% ProofRecord completeness, clean error accounting, recovery with no automatic re-execution of denied requests, and 100% independent signature verification.

Latency/throughput data are explicitly published as production-like overhead characterization, not a benchmark certification and not a production SLA, and must not be compared to the prior in-process microsecond testbed (ARK-483–492).

Standing reconciled corpus across all series to date: 72 preregistered experiments, 232 case records, 229 PASS, 2 FAIL, 1 GATE-STOP, across 9 repositories. Negative results (FAIL, GATE-STOP) are retained unchanged.

**Keywords:** AI agent safety, authorization gate, fail-closed enforcement, proof-carrying decisions, hash chain, ed25519, tamper detection, preregistration, reproducibility

**License:** as per program policy (to be confirmed at deposition).

**Files:** preregistration/ , testbed source tree, proofrecords/proofrecord\_chain.jsonl , ledgers/ , results/ (results ledger, reconciliation report, ARK-498 performance report, execution manifest, verifier outputs).

**Related identifiers:** supersedes preregistration v1.0 (SHA-256 deb9c43ee252ecd9cb217788f783ebf6f-d7113883170749fedaa1509425406ce ).

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### 3. Evidence-page update (executionproof.io / builderexecutionproof.io)

Replace the standing evidence summary block on **both** executionproof.io and builderexecutionproof.io with:

#### Evidence corpus (updated — ARK-493 → ARK-498 series added)

**72 preregistered experiments · 232 case records · 229 PASS · 2 FAIL · 1 GATE-STOP · 9 repositories.**

The latest series (ARK-493–498) added six experiments and 161 scored cases, all PASS, with zero enforcement leaks and independent dual-guard verification on every record. The 2 FAIL and 1 GATE-STOP records from prior series are retained unchanged — we publish negative results because preserving them is what makes the corpus trustworthy.

Highlights: independent tamper detection 10/10 with 0 false positives (ARK-497); fail-closed under dependency loss with leak count 0 and 100% independently signature-verifiable proofs in a production-like networked run (ARK-498). ARK-498 latency/throughput figures are published as production-like overhead characterization — **not a benchmark certification and not a production SLA.**

#### Intellectual-property status (corrected)

ExecutionProof is covered by **48 provisional patent applications (filed January 2026) and 8 pending nonprovisional utility applications — 56 total USPTO filings.**

**Patent-line correction note (for the site editor):** the prior evidence page understated the filing count. The correct, current line — to appear on both executionproof.io and builderexecutionproof.io — is exactly:

48 provisional applications (filed Jan 2026) and 8 pending nonprovisional utility applications = 56 total USPTO filings.

### 4. Consistency checklist before publishing

- [x] Public headline uses reconciled **post-series** totals (72 / 232 / 229 / 2 / 1 / 9), not this-series-only numbers.
- [x] FAIL (2) and GATE-STOP (1) reported, not hidden.
- [x] ARK-498 performance carries the mandatory characterization label and no SLA claim.
- [x] No comparison drawn to ARK-483–492 microsecond testbed.
- [x] Patent line corrected to **56 total USPTO filings (48 provisional filed Jan 2026 + 8 pending nonprovisional utility)** on both domains.
- [x] Preregistration SHA-256 cited and verified at run time.
- [x] Every claimed number traces to a run artifact under `results/` , `proofrecords/` , or `ledgers/` .